GP52: learning corrections to a 1D model

Physical equations compute concentrations. Networks learn information lost by averaging.

Known-inlet prediction

Known chemical inlet
Concentration versus time

→

1D model + closures
Transport and reaction

→

Predicted Th / Fg / Fn
Every position and time

Sparse inlet recovery

Observe Th and Fg
1, 2 or 3 measurement sites

→

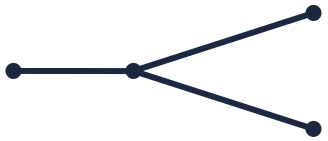
Fit the unknown inlet
Keep the trained model fixed

→

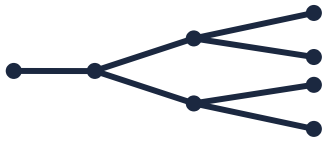
Recover and predict
Inlet and full concentration fields

1 vessel

3 vessels



7 vessels



Current stage: independent references, neural training and fixed-C0 comparisons under one common protocol.

The chemical state and its blood-flow input

One mean concentration per species and axial cell. Blood may vary with position and time.

The simplified reaction

Th promotes the conversion

Fg ⟶ **Fn**

Fibrinogen — Fibrin

Th = thrombin. Th and Fg move.
Fn accumulates locally.

The 1D concentration state

Cell 1	Cell 2	Cell 3
h, g, f	h, g, f	h, g, f

h = mean Th, g = mean Fg, f = mean Fn

Axial cells retain lengthwise variation.
Each value averages a whole cross-section.

$$u(x,t) = Q(x,t) \,/\, A(x,t)$$

A: area. Q: fluid volume per second.
Fixed blood: reuse the same A/Q fields.

Th has no reaction loss. Fn has no axial transport or flow feedback. The chemical state has no radial bins.

Reference data and the 64 / 16 / 32 split

Random inlet conditions produce physically simulated concentrations, fluxes and reaction rates.

Sen 2024 inputs
Geometry and blood conditions

→

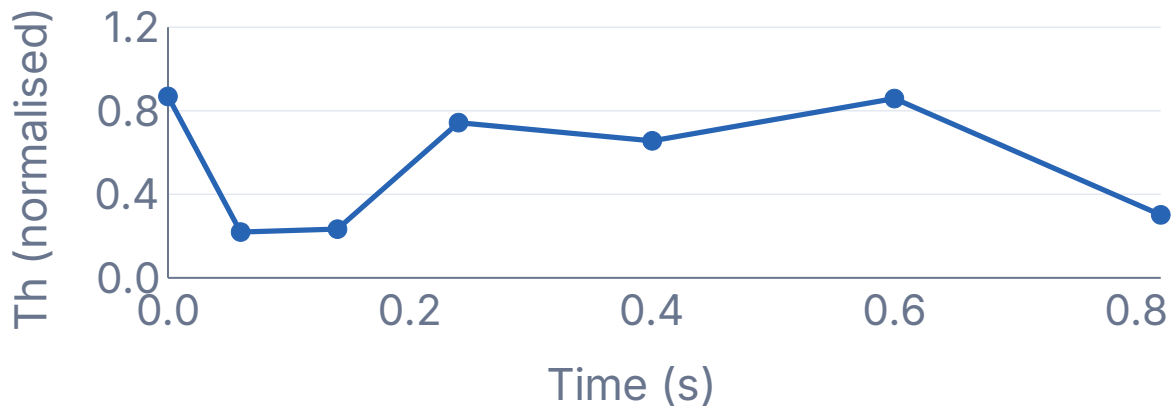
Independent CFD
Blood A, Q and velocity profiles

→

Fine biochemistry
Transport and reaction

→

Average to 1D
Learning and evaluation targets



112 complete inlet histories per topology

7 amplitudes define each Th inlet.
A case includes the full concentration trajectory.

Reference: 24 radial bins, 4× axial mesh, 1 ms steps. Branched blood uses coupled tubes.

64 Train: learn weights	16 Validate	32 Test

Whole trajectories stay in one split. Case 80 is held out. Each topology uses one blood condition.

The reference, C0 and the governing equations

Both candidates solve material balances. We compare their predictions with the finer simulation.

Role	What it provides
CFD reference	Resolved blood and biochemistry for targets and evaluation
Fixed C0	Conventional 1D transport and the fixed reaction law
Neural model	The 1D solver with learned transport and reaction corrections

Storage + transport = reaction

$$\text{Th: } \partial(Ah)/\partial t + \partial J(h)/\partial x = 0$$

$$\text{Fg: } \partial(Ag)/\partial t + \partial J(g)/\partial x = -AR$$

$$\text{Fn: } \partial(Af)/\partial t + 0 = +AR$$

$$J(c) = Qc - AD\,\partial c/\partial x$$

Qc: advection. −AD ∂c/∂x: diffusion.
J: amount crossing a section per second.

$$R = 2hg$$

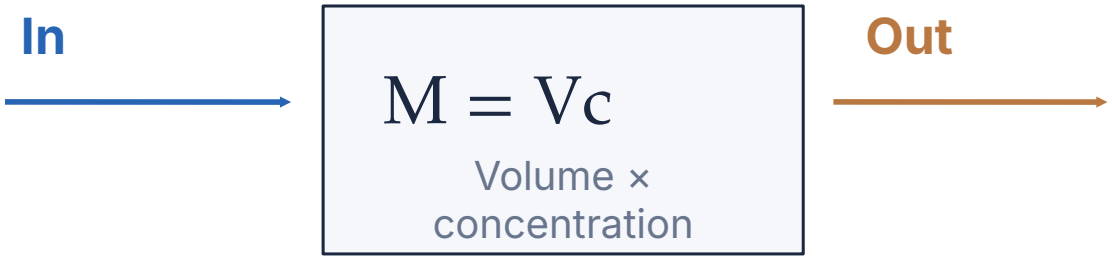
2: benchmark reaction coefficient.
R removes Fg and produces Fn.

c is mean Th or Fg, D is diffusivity. The reference is synthetic simulation data, not patient measurements.

C0 conserves material in cells and at branches

New amount = old amount + inflow − outflow + reaction change.

One finite-volume cell



A zero-volume mixing junction



$$J_p = J_1 + J_2$$

$$M_{new} = M_{old} + \Delta t\,(J_{in} - J_{out}) + \Delta t\,V_{new}s$$

Δt: step length. s: reaction source (0 for Th, negative for Fg, positive for Fn).

Update Th
Transport

→

Update Fg
Transport + consumption

→

Update Fn
Matching reaction production

2 ms implicit steps solve linked concentrations. Junctions balance total flux, so concentration is not halved.

Why cross-sectional averages need closure

Averaging loses spatial correlations in both reaction and transport.

Reaction: where Th and Fg overlap

Equal zones	Th	Fg
A	1.2	0.8
B	0.8	1.2

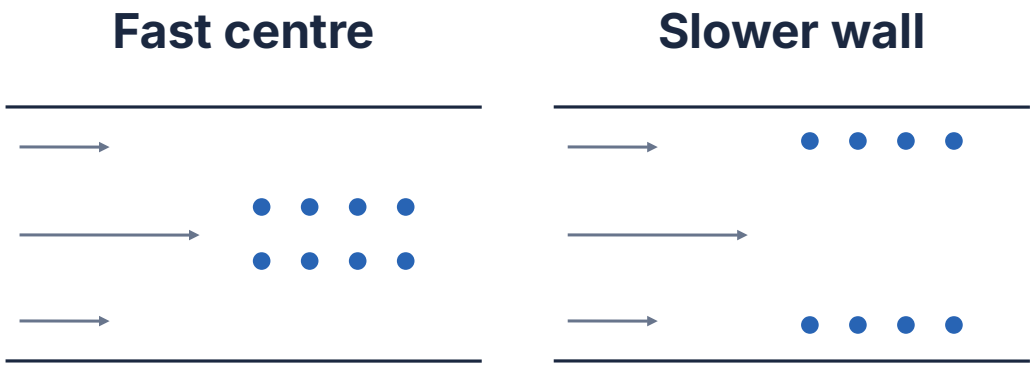
React using means

$$2 \times 1 \times 1 = 2.00$$

Average the local rates

$$2 \times (0.96 + 0.96)/2 = 1.92$$

Transport: where solute meets flow



$$\text{mean}(uc) \neq \text{mean}(u)\ \text{mean}(c)$$

Same mean concentration and flow.
Different distributions can change flux.

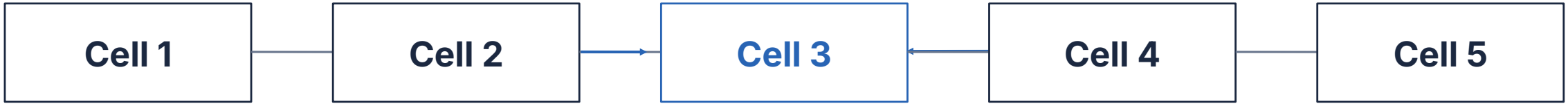
Teaching examples. More accurate numerical integration of C0 does not recover the missing correlations.

The neural model retains physics and uses neighbours

Networks output closure coefficients. The conservative solver computes concentrations.

Component	Fixed C0	Neural model
Mean state	One Th / Fg / Fn value per cell	One Th / Fg / Fn value per cell
Transport	Advection + diffusion	Same terms + memory flux
Reaction	2hg	2κhg with learned κ
Junctions	Conservative mixing	Conservative mixing

A graph links neighbouring computational positions

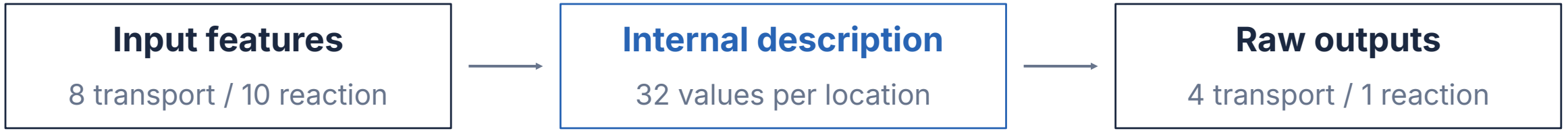


Cell 3 combines itself with Cells 2 and 4. Transport uses faces, reaction uses cells.

Separate heads and topology weights. Two scalar memory states supplement 1D concentrations, without radial bins.

Graph computation and transport coefficients

The two heads use the same network form, with separately trained weights.



$$z_{new} = \tanh[z + 0.25(W_c z + b + W_n m)]$$

Repeat twice. z: local features. m: neighbour mean. W / b: learned weights / bias.
tanh: a bounded nonlinear transform.

Transport inputs: 8 features

Position and spacing (2), area and radius (2), current velocity (1), filtered velocity histories (3).

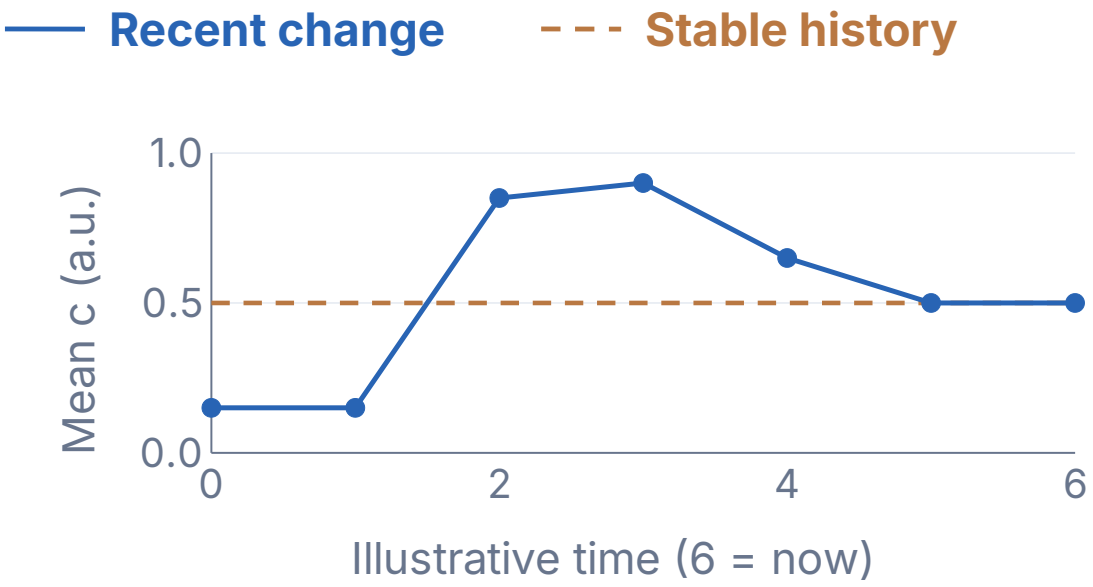
$\alpha_1, \tau_1, \alpha_2, \tau_2$

α: coupling strength
τ: memory duration

Linear maps enter and leave the hidden space. 32 features are internal numbers. Single-vessel coefficients differ.

Transport memory: motivation and update

The same present mean concentration can follow different histories and radial distributions.



Radial redistribution takes time.
Memory retains part of that history.

Decayed old memory + gradient response

$$m_{new} = d\, m_{old} - \tau(1-d)\alpha G_{new}$$

d = exp(−Δt/τ): retained fraction
G: axial concentration gradient

$$J_{memory} = A(\alpha_1 m_1 + \alpha_2 m_2)$$

Add this flux to advection and diffusion.

Teaching curves. 3 / 7 vessels use local memory recurrence. The single-vessel model also advects memory.

Reaction correction and the complete prediction

Ten reaction features use predicted Th/Fg, gradients, blood flow and recent history.

$$\kappa = \exp[0.25 \tanh(z)]$$

$$R = 2\kappa hg$$

z: raw network output. κ is positive, about 0.78–1.28. κ = 1 gives C0.

FIRST PASS: obtain the state needed by the reaction network

Transport GNN
Blood and geometry give α, τ

→

Predictor solve
Transport correction, κ = 1

→

Predicted state
Th, Fg and their gradients

SECOND PASS: use the learned reaction rate

Reaction GNN
Predicted features give κ

→

Corrected solve
Transport and reaction closures

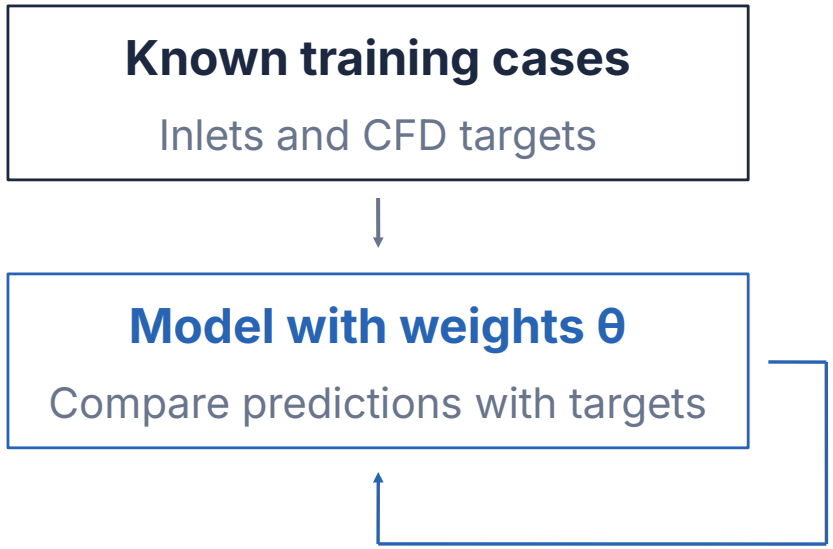
→

Final concentrations
Th / Fg / Fn over space and time

κ < 1 weakens reaction, κ > 1 strengthens it. The same corrected rate removes Fg and adds Fn.

Training: weights, losses and update counts

Known inlets and CFD targets train θ. Validation selects checkpoints. Test cases assess the final choice.



Loss drives updates of θ

Train transport
Select and freeze transport
Train reaction

Three / seven: normalised physical-target losses

$$L_{flux} = MSE(J_\theta, J^*)/S_J$$

$$L_{rate} = MSE(R_\theta, R^*)/S_R$$

J: extra flux. R: rate. *: reference. S: reference mean-square scale.

Single: concentration and reaction objectives

$$L_{transport} = E_{curve} \qquad L_{reaction} = E_{curve} + wE_{rate}$$

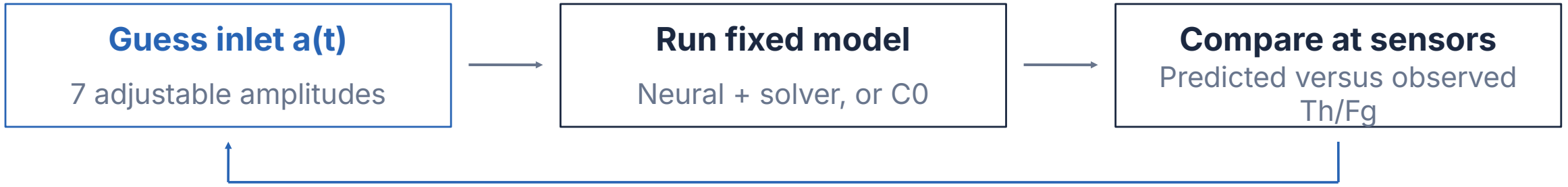
Curve: Th/Fg/Fn for transport, Fg/Fn for reaction.
Rate: reaction-rate error. w: validation-selected balance.

Training stage	Updates
Single transport	1,500
Single rate pretrain + mixed refine	1,000 + 500
Three / seven: each head	3,000

MSE averages squared differences. Seeds 42 / 43 / 44. Single reaction weights: 0.03 / 0.3 / 3.

Sparse inversion adjusts the inlet, with fixed weights

Observe Th/Fg at 1, 2 or 3 sites. Each site supplies 24 times × 2 species = 48 noisy values.



Update the inlet a. Keep network weights θ fixed.

$$\min_a [\, E_{data}(a) + \lambda E_{smooth}(a) \,]$$

$$E_{data} = \|\hat{y}(a) - y_{obs}\|^2 \,/\, (48s)$$

ŷ: predicted sensor values. y_{obs}: observations.
s: number of sites.

$$E_{smooth} = \|La\|^2 \,/\, 5$$

La: five changes in inlet slope.
Larger λ favours a smoother inlet.

Each method chooses λ on validation and fixes it for testing. Fn stays hidden and is evaluated afterwards.

Evaluation protocol

C0 and the neural candidate face the same test cases and inverse observations.

Shared inputs and inverse rules

Same blood, observation sites, noise and inlet bounds

Same 3 starting rules. Each method tunes smoothing on validation.

What an error percentage means

$$\text{Relative } L_2 \text{ error} = \text{size of prediction error} \,/\, \text{size of reference} \times 100\%$$

0% means an exact match. Lower is better.

Average over 32 test cases. Score inlet to 0.82 s and concentrations to 1.23 s.

Prediction accuracy when the inlet is known

Mean relative L_2 error (%) over 32 unseen inlet cases per topology.

Each entry compares **C0** → **Neural**

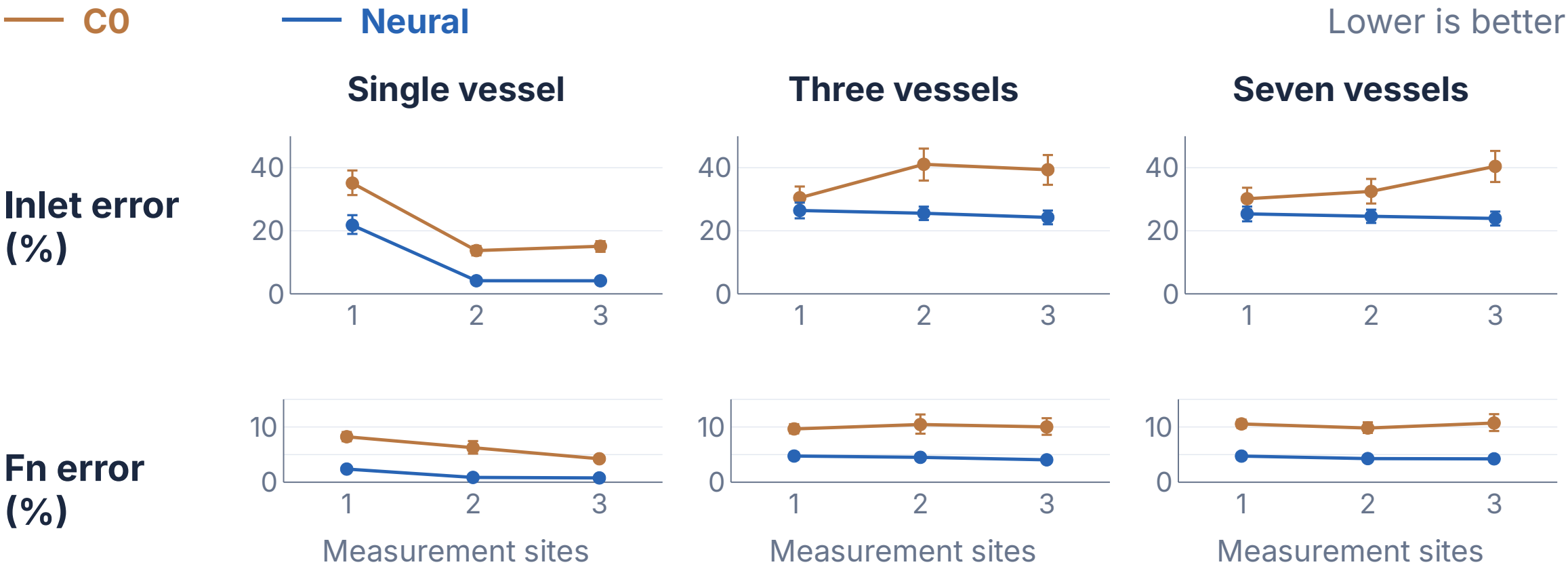
Network	Th	Fg	Fn
Single vessel	14.09 → 1.41	5.31 → 0.60	8.27 → 0.52
Three vessels	13.56 → 4.92	5.22 → 1.85	8.90 → 2.74
Seven vessels	14.19 → 4.91	4.60 → 1.60	9.65 → 2.78

The neural model has lower mean error in all nine comparisons.

These are whole-network averages. Individual locations and cases can behave differently.

Inverse accuracy with one, two and three sites

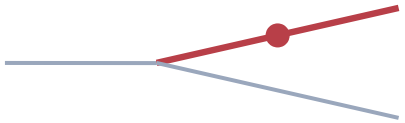
Mean errors with 95% case-bootstrap intervals. Fn is never an inverse observation.



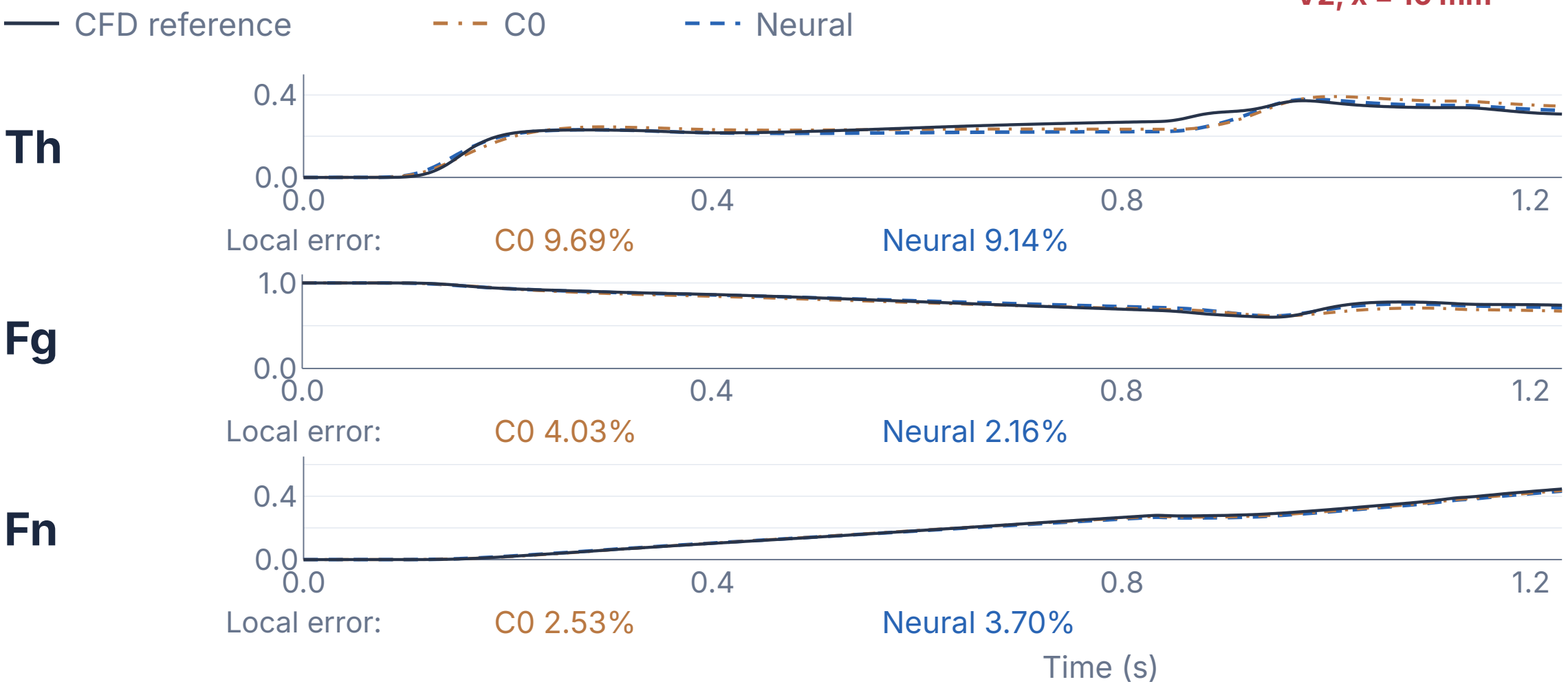
Each method uses the same three starting rules and its own validation-selected regularisation.

Example: concentrations in a daughter vessel

Normalised concentrations. Case 80 with a known inlet.



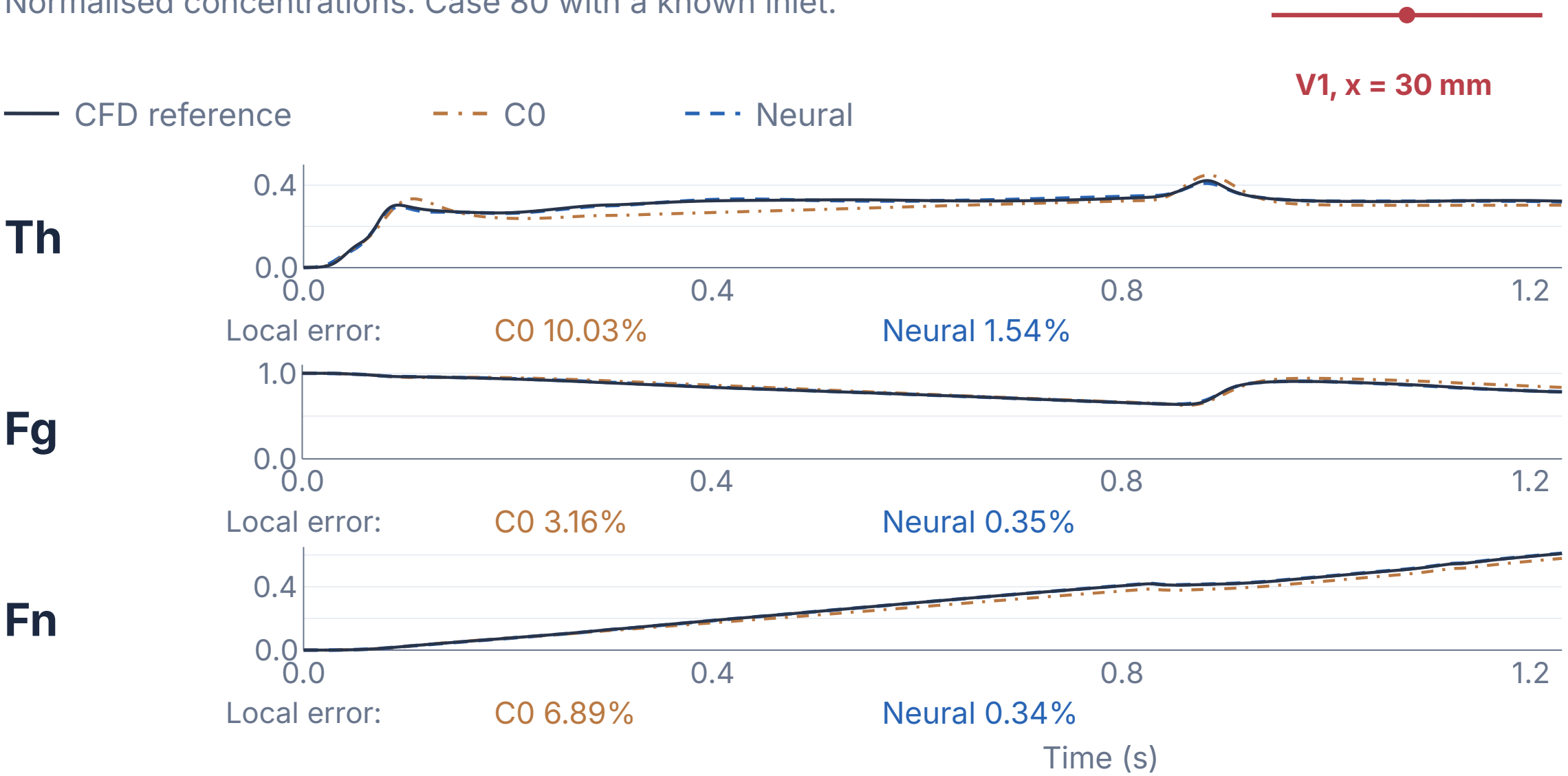
V2, x = 10 mm



616 samples, neural seed 42. Local errors for one example, distinct from the 32-case average.

Single vessel: concentration changes over time

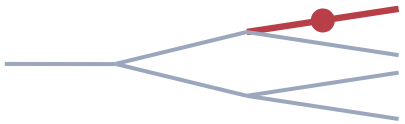
Normalised concentrations. Case 80 with a known inlet.



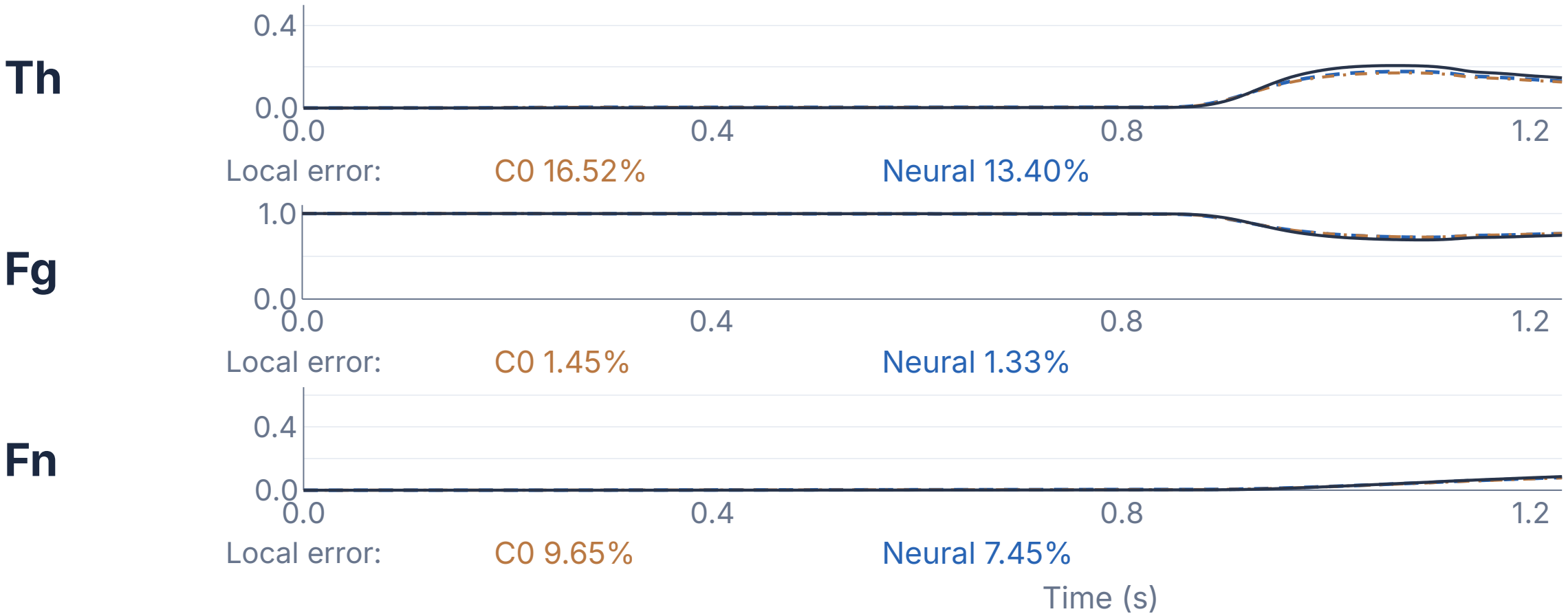
616 samples, neural seed 42. Local errors for one example, distinct from the 32-case average.

Seven vessels: concentration changes over time

Normalised concentrations. Case 80 with a known inlet.



V4, x = 19.5 mm



616 samples, neural seed 42. Local errors for one example, distinct from the 32-case average.

Three-site reconstruction and the scope of the result

Mean relative L_2 error (%) over 32 held-out cases. Each entry compares C0 with the neural model.

Network	Inlet	Th	Fg	Fn
Single vessel	15.08 → 4.14	8.41 → 1.87	3.40 → 0.60	4.24 → 0.78
Three vessels	39.37 → 24.25	14.89 → 7.24	3.74 → 1.60	10.00 → 4.05
Seven vessels	40.43 → 23.91	15.25 → 6.94	3.21 → 1.40	10.71 → 4.23

Demonstrated under this protocol

Lower mean prediction and reconstruction errors.
Fn improves although it was hidden during fitting.
Branched inlet errors remain about 24%.

Evidence still needed

New blood conditions and geometry.
Speed and graph-specific benefit.
Local exceptions remain (see the V2 example).

Concentration errors cover the full network. Accuracy gains come from learned closure information in the solver.

Current limitations and next evidence

Three observed limitations and one untested capability in the current unified evaluation.

Limitation	Neural-model evidence	Next check
Inlet recovery remains difficult	3 sites, no noise Three: 22.37% Seven: 22.18%	Separate model error, observability and fitting.
Modest gains from more measurement sites	1 site vs 3 sites, with noise Three: 26.44% to 24.25% Seven: 25.36% to 23.91%	Test sensor positions and inlet sensitivity.
Residual transport error	Known-inlet Th error Three: 4.92% Seven: 4.91%	Isolate transport from reaction corrections.
Generalisation remains untested	One blood condition per topology. Separately trained weights.	Freeze weights and test new flow and geometry.

Further checks: seven-vessel reference convergence and equal-parameter no-graph ablation.
Meeting decisions: closure focus, experiment priority, success criteria and team roles.

Errors: mean relative L_2 over 32 test cases. Sensor distances, geometry and flow differ across topologies.